EXP 5:

Software Requirements Prioritization using MoSCoW Analysis for Library Management System

Aim:

To analyze, prioritize, and categorize the functional requirements of a Library Management System using the MoSCoW method based on user impact and feasibility constraints.

procedure:

• Open Spreadsheet: Launch Google Sheets or MS Excel and create a new blank sheet. • Add Headers: Set Row 1 headers as: Req ID, Requirement Description, MoSCoW Category, Impact on Users, and Feasibility & Justification.

• List Requirements: Enter all 10 library system requirements into the table.

• Categorize Requirements: Assign each requirement to Must-Have, Should-Have, Could-Have, or Won't-Have based on operational need and feasibility.

• Add Justifications: Write a short explanation for each item detailing its user impact and technical effort.

• Format & Export: Apply bold headers, auto-fit columns, add text wrapping, and save or download as .xlsx or PDF for submission.

Output:

B12

Req ID	Requirement Description	MoSCoW Category	Impact on Users / Stakeholders	feasibility	Resource Justification
REQ-01	Add a feature to search books by title and author.	Must-Have	high	high	low complexity, low cost; essential for baseline deployment.
REQ-02	Implement an online book reservation system.	Must-Have	high	high	standard database transaction logic requiring minimal budget.
REQ-03	Generate monthly reports on borrowed books for administrators.	Must-Have	high	high	standard SQL database query and reporting engine function.
REQ-04	Enable email notifications for overdue books.	Should-Have	high	high	using standard SMTP mail services with low integration overhead.
REQ-05	Create a user-friendly dashboard for librarians.	Should-Have	high	medium	requires UX design and backend API endpoints.
REQ-06	Add support for QR code scanning for borrowing and returning books.	Could-Have	medium	medium	requires webcam/mobile integration and hardware testing.
REQ-07	Allow users to review and rate books.	Could-Have	medium	high	basic CRUD functionality, but non-essential for initial launch.
REQ-08	Provide multi-language support.	Could-Have	medium	medium	requires i18n localization setup and translation management.
REQ-09	Integrate a chatbot for user assistance.	Won't-Have	low	low	high complexity, cost, and AI integration efforts for initial release.
REQ-10	Develop a mobile app version of the system.	Won't-Have	medium	low	requires cross-platform development, time, and team budget.

Result:

Thus, All 10 requirements for the Library Management System were successfully analyzed, categorized, and documented in Google Sheets/Excel using the MoSCoW method. Each requirement was evaluated based on user impact and technical feasibility to establish a clear development priority list.